Ensemble Approach to Classify Spam SMS from Bengali Text

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Abstract The Short Message Service (SMS) is a popular communication tool, but it has some security weaknesses, such as the influx of spam messages from cyber criminals. While several studies have been conducted on filtering and categorizing spam messages in various languages, including English, limited research has been done on detecting spam in Bengali (endonym Bangla) text. This study aims to fill this gap by classifying Bengali SMS messages as either spam or ham (legitimate messages). To accomplish this, the study used machine learning algorithms, including support vector machine (SVM) with a linear kernel and decision tree (DT), logistic regression (LR), and random forest (RF) with various parameters, as baseline models. Ensemble approaches, such as bagging, boosting, and stacking, were then used to enhance the performance of the models. The results show that the ensemble approach successfully identified spam messages in Bengali text, with XGBoost producing the most favorable outcome. The contribution of this study lies in its focus on Bengali text and the demonstration of the ensemble method's performance on a small dataset. The tool developed in this study can provide a secure and efficient SMS service to customers by reducing the burden of spam messages and improving the overall user experience. Additionally, the tool can be marketed as a value-added service for customers who are concerned about the security of their personal and financial information. Overall, this study highlights the importance of machine learning algorithms, specifically ensemble methods, in detecting spam messages in Bengali text and provides a valuable contribution to the field of SMS security.

Keywords: Bengali Text · Machine Learning · Ensemble Method · SPAM SMS · Ensemble · Classification.

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1 Introduction

Globally, a short messaging service (SMS) is one of the most popular and affordable telecommunication service packages [20]. It is the most convenient mode of communication for the young generation. Nowadays, SMS marketing is an effective way for businesses to reach their customers, as it allows for quick and direct communication. However, business companies must ensure they have obtained proper consent from their customers before sending any marketing messages via SMS. This is mainly because mobile marketing remains intrusive to the personal freedom of the subscribers [8]. Because SMS allows for immediate, direct engagement with clients, it can be an effective tool to inform or advertise. More or less, everyone in every country prefers to trust SMS over any online-based messaging service [7].

Unwanted messages can be sent electronically under the term “spam”. While SMS messages are conveyed through the mobile network, spam e-mails are sent over the internet [11]. Additionally, spam SMS poses security risks since they may contain links that take consumers to malicious websites [7]. SMS marketing’s popularity can be a target for cybercriminals, who may use it to send spam or unsolicited messages to the receivers to get personal information or access dangerous websites. This will force them to divulge their private information to an unreliable source. Spammers may also use this information for targeted phishing attacks or to sell to third parties. The trust in SMS service is advantageous for spammers [5]. Additionally, it is quite unpleasant for the recipients. Which intentionally raises the alarm about personal information [17].

“The spam text” problem can be solved by employing more complex SMS filtering techniques, such as machine learning algorithms, to recognize and prevent spam texts automatically. This can help to limit the number of spam messages that reach the customer’s inbox, making it easier for them to identify relevant information. That kind of classifier can be produced via a machine learning-based algorithm [2]. It is crucial to note that SMS filtering techniques, including machine learning algorithms, are not flawless. There is always the risk of false positives (genuine messages being labeled as spam) or false negatives (illegal messages being marked as spam). Therefore, a new generation of intelligent security systems is being launched for data filtering [4]. It is challenging to filter datasets word-by-word in another language. Outdated models cannot control the escalating number. Therefore, new algorithms may be efficient and require less work. So a mechanism for SMS filtering and spam SMS detection is created by this work.

However, it is important to note that developing a system capable of accurately identifying SPAM SMS written in Bengali (endonym Bangla) may be difficult. There may not be a standardized classification system in place, and the task may necessitate a significant amount of data and resources. Furthermore, the system would need to be updated and modified regularly to keep up with the changing nature of spam messages. There are several works to detect SPAM SMS from English text. But, Bengali text SMS currently lacks a consistent classification system, making distinguishing between SPAM messages challenging. It will take a lot of data, resources, and knowledge in natural language processing (NLP) and machine learning to develop a system recognizing spam SMS. According to Google, Bengali is the seventh most widely-used language in the world. Many people use this language. However, some work has been done based on the Bengali language, and the outcomes of that work are not efficient. So, SPAM SMS filtering from Bengali text imposes some challenges. Due to this, we propose a system that can differentiate between spam and ham SMS from Bengali text using ML and NLP techniques. For classifying the SPAM SMS, we manually collected the dataset.

2 Related Work

Several initiatives have been completed in this sector, and this chapter showcases some of them. By examining these initiatives, we can identify their limitations and pinpoint areas where additional efforts are required. In the subsequent sections, we will explore the relevant completed works and assess any shortcomings or gaps that were encountered.

The analysis of message filtering was conducted in a study [12], where the authors employed dual filtering, K-Nearest Neighbor (KNN) classification, and rough set techniques. They proposed the concept of dual-filtering communication and used the primary classification techniques of Naive Bayes (NB), Support vector machine (SVM), and KNN. To achieve high accuracy and efficiency in their new method, the KNN algorithm re-filters communications previously classified as spam. In [23], researchers conducted a study describing the Fake-base-station Spam Ecosystem in China. They provided a comprehensive overview of the FBS spam environment by collecting three months' worth of actual FBS detection results. Over this period, they gathered FBS messages from spammers in China and acquired supplementary datasets to perform an in-depth analysis of the FBS ecosystem. The study found that contacts were evenly distributed across campaigns, with over 96% of campaigns involving a single connection for all messages. The effectiveness of several evolutionary learning classifiers for filtering mobile spam (SMS) was analyzed in a study [16] using a real-world dataset by researchers. The aim was to identify the best classifier for spam detection from a large pool of evolutionary and machine-learning classifiers. The study found that the supervised classifier system (UCS) produced a detection rate of more than 89% and a false alarm rate of 0% when used with the specified features. A trust-based management system for controlling SMS spam was developed and applied, as presented by the authors in a study [9]. They provided the prototype concept and implementation of the system, which demonstrated excellent accuracy, efficiency, and resilience for SMS messages on mobile devices. The study showed the positive commercial potential of the GTM system and its importance in addressing the growing issue of undesirable traffic lights (UTC) in communication networks.

SMS Classification: SMS classification is the process of sorting documents into predefined classes based on their content. The researchers in [11] use Word2Vec feature extraction to analyze SMS classification. In another research by Zhang et al. [22], an empirical comparison of English and Chinese spam text classification was made using publicly available Internet datasets. Information Gain (IG) IG-KNN has the highest accuracy and stability of classification among the four classifiers examined; however, it performs poorly when classifying Chinese.

Spam SMS Detection: Spam SMS detection is the process of identifying unwanted SMS and filtering them out of the inbox. Research is being done on filtering external mobile SMS spam by Androulidakis [5], and Merugu [17]. A weighting technique based on the propensity of words to indicate specific target classes (HAM or SPAM) has been employed to categorize new SMSs as SPAM or HAM in a study [17]. The False Positive (FP) to False Negative (FN) ratio is widely used in epidemiological studies as it plays a vital role in evaluating the effectiveness of various security software and systems.

Dual Filtering Message: Dual filtering of messages refers to using two different techniques to filter and process messages to improve the accuracy and reliability of that system. Another study [12] mentioned using dual filtering, KNN classification, and rough set techniques for analyzing message filtering. The concept of dual-filtering communications is suggested in the study. To achieve high classification accuracy and speed in the new method, the KNN algorithm re-filters these communications previously classified as spam.

Language-based Spam SMS Detection: Due to the variation in speech architecture across different languages, it is essential to develop a new algorithm for detecting SMS spam for each language. Turkish spam sms was detected by Onur (Spam SMS Detection for the Turkish Language with Deep Text Analysis and Deep Learning Methods) compared to Chinese and English spam text by Liumei [23]. Theodorus et al. created a dataset of Indonesian SMS used in their investigation [21]. In addition, they did categorize SMS into three categories: spam, ham, and promotional texts. The training was done in two clumps utilizing different dataset proportions. The classifiers were evaluated using a 10-fold cross-validation metric. The findings indicate that Extreme Gradient Boosting (XGBoost) 94.52%, Multinomial Logistic Regression (MLR) 94.57%, SVM 94.38%, and Random Forest (RF) 94.62% are some of the top models for a multiclass SMS classification.

Fake-base-station Spam Ecosystem: A description of "The Fake-base-station" is given by Chinese scholars [23]. This paper presented the first large-scale depiction of the FBS spam ecosystem by collecting three-month certifiable FBS discovery results. They managed FBS messages sent by genuine spammers in China over three months and gathered a large number of crucial secondary datasets to conduct an in-depth analysis of the FBS environment. Contacts are distributed more evenly throughout campaigns, with more than 96% of campaigns utilizing a single connection for all communications.

The literature reviewed above highlights the existing gaps in the field, particularly the significant challenge of detecting SPAM SMS from Bengali text.

Examining these works further reveals the limitations and gaps in the existing research work.

3 Methodology

The proper steps lead to the highest accuracy of a model. So, we followed some steps to get our results accurately and effectively. We decided to break it down into several stages. Data collection, labeling, preprocessing, feature extraction, model training, and outcome analysis are the stages. Figure 1 depicts the workflow for the suggested approach.

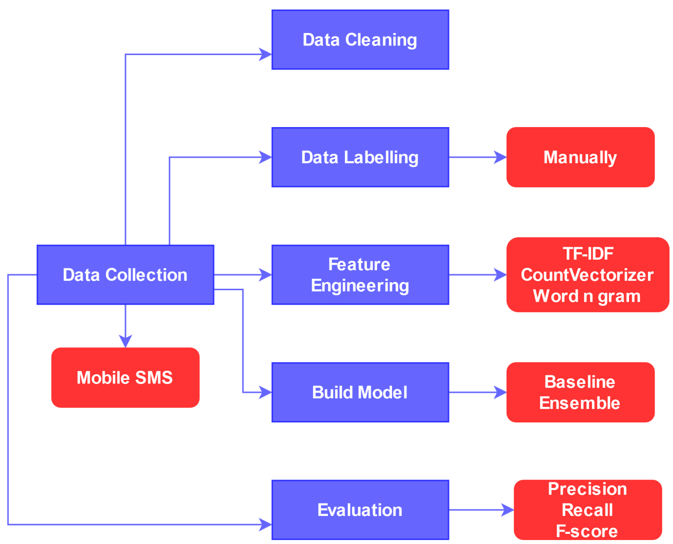

Figure 1. Workflow of the proposed system.

3.1 Dataset Preprocessing and Labeling

The raw text data must undergo some preparation before being sent to the classifier [3]. Firstly, we clean the data. In the raw dataset, there can be numbers, emojis, hashtags, URLs, etc., which can impact the outcome of the system. So, data cleaning is a necessary step for this research. In these steps, we perform various activities. The actions performed in the data cleaning part are discussed below:

The Python library was used to remove the number, digits, emoji, and URLs. However, we do not use hashtags in our situation. We also employ additional standard text preparation techniques, such as punctuation removal, white space removal, accent mark removal, stop word removal, etc. Unlike the English language, Bengali does not have a designated stop word. This is dependent on a particular task. We utilized the Python library and regex to remove unnecessary words and punctuation from the text after carefully identifying all stop words that did not include objectionable language. We collected the SMS from each of our phones. The SMS is slated to launch between 2020 and 2022. Only SMS written in Bengali is taken into consideration. We manually labeled every message in our dataset after it was assembled. We classified them into two groups (Spam, Ham). The number of each label is shown in Figure 2. The expert in this field, who has experienced more than five years and has depth knowledge of the telecommunication sector, was the supervisor of data labeling. Some samples in the Bengali SMS dataset that we used are demonstrated in Table 1.

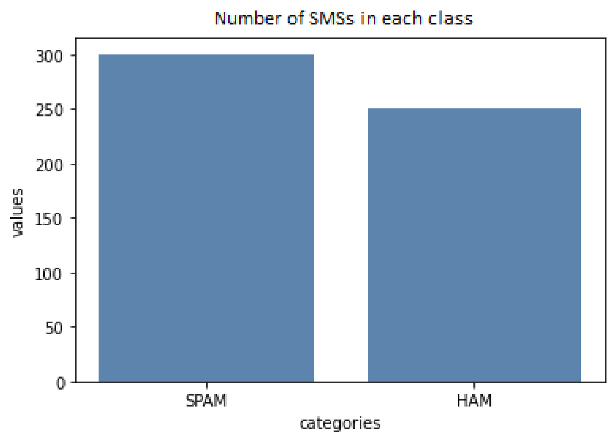

Figure 2. Number of SMSs in two classes of dataset.

3.2 Feature Extraction

In the context of text classification, each model requires the input of a numerical vector that represents a text document. The process of converting the words in a text document into fixed-length floating-point or binary vectors is referred to as feature extraction or vectorization. Our research focuses on discussing the techniques used for feature extraction, as outlined below.

TF-IDF Vectorizer: The TF-IDF technique calculates the weight of a word by considering both its frequency (TF) within a document and the number of documents in which it appears (IDF). This allows for each word's weight to be effectively determined as the IDF takes into account the presence of a term across multiple documents [15].

Count Vectorizer: The bag-of-words approach is used by the count vectorizer to avoid textual structures. To extract information, just word counts

Table 1. Dataset example.

Example	English Translation	Data Label
১৫০ টাকায় ২৫ জিবি ইন্টারনেট নিতে ডায়াল করুন	Dial for 25GB internet at 150 Taka.	HAM
সুঃ খবর,আপনি গ্রামীনফোনের লটারি নির্বাচনে জয়ী হয়েছেন।এখনি আপনি লিংকে প্রবেশ করন।	Good news, you have won Grameenphone lottery selection. Now you enter the link.	SPAM
প্রিয় গ্রাহক, রিন ১ কেজি পাউডার কিনলেই পাচ্ছেন ২০ টাকা ছাড়! আজই কিনুন আপনার নির্দিষ্ট চক বাজার থেকে; অফারটি পেতে বাজারে অবস্থিত লাভের বাজার বুথে চলে আসুন। অফার চলবে ১১ই জানুয়ারী থেকে ২১ই জানুয়ারী পর্যন্ত।	Dear customer, you can get 20 taka discount if you buy Rin 1 kg powder! Buy today from your specific Chuck Bazaar; Visit the Love Bazaar booth located in the market to avail the offer. Offer valid from 11th January to 21st January.	HAM
খণ্ডকালীন জব।বেতন 1000 BDT1/দিন।ঘরে বসে কাজ :wa.me/8801917379501	Part Time Job Salary 1000 BDT1/day Base work :wa.me/8801917379501.	SPAM

are used. Each document will be converted to vector format. The input to the vector is a count of how many times each particular word appears in the material [10]. The n-grams and count vectorizer combinations approach works best in our research.

Word N-grams: The N-grams feature is thought to be quite effective for text classification. This consists of the next n words or characters. We employed the word N-grams in our work. To find the optimal model, we examined the performance of unigrams, bigrams, and trigrams. To achieve the overall scenario, we integrated the N-grams feature with other features [6]. We used Scikit-learn to experiment with features [19].

The count vectorizer, TF-IDF vectorizer, and word N-grams approach mentioned above were used for feature extraction. The best results for our model are obtained by combining word N-grams and TF-IDF vectorizer. TF-IDF converts text into a vector that can be used. It increases a system's precision. The N-grams feature is also regarded as the best for text classification. N-gram features (2,3) were used to achieve the best possible result.

3.3 Classification Model

Data collecting and preprocessing are the initial part of our research. After that, to fit the data into the model, data were converted into vector formats by applying many feature extraction techniques. Finally, the model was built up. In our research, the models that are involved are shown here.

Baseline Models: We first used the baseline models to categorize the SPAM and HAM SMS. To categorize the SMS, we used support vector machine (SVM) with a linear kernel. Then, we applied decision tree (DT), logistic regression (LR), and random forest (RF) with various parameters.

When using the baseline model, we applied several parameters. Table 2 shows the parameters used for different baseline classifiers and their values.

Table 2. Baseline algorithms details (parameters).

Algorithm	Parameters	Values
DT	criterion, max_depth, max_features	entropy, 5, 10
RF	n_estimators, criterion, max_depth	100, entropy, 3
LR	penalty, class_weight, random_state, max_iter	l1, balanced, none, 200

The baseline algorithm produced unsatisfactory results. As a result, we choose to use some of the other models. Due to the ensemble approach's ability to mix two or more models, we used it. Recently, Al Maruf et al. [1] used an ensemble model to classify the text, and then the accuracy was increased. We used bagging, boosting, and stacking models to detect SPAM and HAM SMSs.

Bagging: It stands for Bootstrapped Aggregation, is a machine-learning ensemble technique that involves aggregating the results of multiple base estimators to produce a more robust and accurate final prediction [13]. We applied different base estimator models, but When the base estimator is a LR model, and the n_estimators value is set to 100, it performs best. In this setup, 100 instances of the LR model are trained on different randomly sampled subsets (with replacement) of the original training data. Each of these 100 models makes predictions on a new data point independently. The final prediction is obtained by taking a majority vote (in the case of a classification problem) or by averaging the outputs (in the case of a regression problem). By aggregating the predictions of multiple models, the Bagging Ensemble Model can reduce the variance of the predictions and produce a more accurate final prediction compared to a single Logistic Regression model. Moreover, since each base estimator is trained on a different subset of the training data, the Bagging Ensemble Model has the ability to capture different aspects of the underlying data distribution, further improving the accuracy of the final prediction. In this case, it appears that a Bagging Ensemble Model with 100 LR base estimators performs better compared to using a different value for n_estimators. Bagging output depends on several parameters. We used different classifiers as base estimators in our work to get the best result. Firstly, we created the object of each baseline algorithm we used. Then each object was used as a base estimator. Finally, others parameters values were applied differently to get the best result. Finally, LR performs best as a base estimator. Other parameters used in our works and their values are (n_estimator=200, max_samples=0.8). When the n_estimator value was increased, the performance of the Bagging classifier was improved.

Boosting: The intention was used to bagging algorithm to increase the accuracy. But we found some training errors in our model. For this reason, To decrease training errors boosting model was used. It transforms a collection of weak learners into strong learners [14]. Boosting is an iterative technique where each iteration focuses on correctly classifying the data that was misclassified in the previous iteration. The final prediction is a weighted average of all the predictions made by the individual models. Boosting algorithms, such as Adaboost and XGBoost, are specifically designed to address the issue of overfitting.

Adaboost (Adaptive Boosting) assigns higher weights to the instances that were misclassified in previous iterations and train the base model on these instances to produce a better prediction. This helps to focus the model's attention on the more complicated samples, resulting in a more robust model.

XGBoost (eXtreme Gradient Boosting) is an optimized implementation of gradient boosting. It uses a tree-based model as the base model and trains the model using gradient descent to minimize a loss function. XGBoost also includes several techniques to prevent overfitting, such as regularization and early stopping, which help to control the complexity of the model and prevent it from overfitting to the training data. In summary, as the bagging model is facing overfitting issues, we applied boosting techniques like Adaboost or XGBoost to overcome these issues and improve the model's performance. Table 3 displays the Adaboosting and XGboosting parameters.

Table 3. Boosting ensemble parameters.

Adaboosting	**XGboosting**
base_estimator (RF)	base_estimator (DT)
n_estimators (350)	n_estimators (500)
max_samples (0.5)	max_samples (0.8)
-	learning_rate (0.2)

Stacking: It is a type of ensemble learning technique in machine learning that combines the predictions of multiple models to produce a more accurate prediction. It works by training multiple base models on the same training data and using their predictions as input features to train a higher-level model, known as the meta-model. The final prediction is made by the meta-model [18].

The main idea behind stacking is to leverage the strengths of different models and to produce a more robust and accurate model by combining their predictions. The base models can be of different types, such as DT, SVM, etc. Stacking can be performed in two ways: (i) level-0 models: Each base model makes its own independent prediction. These predictions are then combined with the meta-model to produce the final prediction, and (ii) level-1 models: The outputs of the base models are used as input features to train another model, which makes the final prediction. This second-level model is referred to as a meta-model. In our research, we used different algorithms as a base model and meta models. Finally, the base algorithms that performed well in our research when we used DT, RF, and KNN, and the meta classifier is LR.

4 Result and Discussions

As mentioned in the section above, we used bagging, boosting, and stacking approaches. First, we compute the accuracy of each model. The feature extraction techniques count vectorizer, word N-grams, and TF-IDF was utilized, with the combination of TF-IDF and word N-grams producing the best results.

We used a train-test split for model selection. From the dataset, 80% were randomly selected (with stratification) for training and the rest 20% of the data for testing.

The baseline models' performances are shown in Table 4. The training accuracy score indicates the accuracy of the model on the data it was trained on. For instance, the decision tree algorithm has a training accuracy of 76.54%, meaning that it correctly predicts the output for 76.54% of the training data. The testing accuracy score is the accuracy of the model on new, unseen data. For example, the decision tree algorithm has a testing accuracy of 55.98%, meaning that it correctly predicts the output for 55.98% of the new data. The training accuracy is generally higher than the testing accuracy because the model may overfit the training data, memorizing the training data instead of learning the underlying patterns. This can result in a lower accuracy on new, unseen data. In this case, Logistic Regression has the highest testing accuracy of 60.97%. This means that it has the best accuracy in predicting the output for new, unseen data among the four algorithms. However, it's important to note that accuracy is not the only measure of model performance. Other metrics such as precision, recall, and F1-score may also be necessary depending on the problem you are trying to solve. Typically, the result is not up to the mark.

Table 4. Accuracy of baseline models.

Algorithm	Training accuracy	Testing Accuracy
DT	76.54%	55.98%
RF	72.96%	54.97%
LR	77.54%	60.97%
SVM	73.54%	62.65%

To achieve better results, we applied the ensemble method and calculated all of the evaluation metrics. Table 5 shows the testing accuracy and training accuracy scores for four different ensemble models: Bagging, Adaboosting, XG-boosting, and Stacking. Bagging has a testing accuracy of 75.89% and a training accuracy of 99.74%. This indicates that the model is overfitting to the training data, as the training accuracy is significantly higher than the testing accuracy.

Adaboosting has a testing accuracy of 79.76% and a training accuracy of 92.36%. This is a better result compared to Bagging, as the difference between the training and testing accuracy scores is smaller. Adaboosting is another ensemble method that trains multiple base models, but it assigns different weights to the training samples based on their prediction errors. XGboosting has a testing accuracy of 83.87% and a training accuracy of 90.32%. This is a good result, as the testing accuracy is high, and the difference between the training and testing accuracy is not too extensive. XGboosting is an implementation of gradient boosting, an ensemble method that sequentially trains multiple base models, each model attempting to correct the errors of the previous model. Stacking has a testing accuracy of 82.65% and a training accuracy of 94.76%. This is a good result, similar to XGboosting. Stacking is an ensemble method that trains multiple base models and uses their predictions as input features to train a higher-level model that makes the final prediction. Overall, XGboosting and Stacking have the highest testing accuracy among the four models. The bagging, Boosting and stacking ensemble model performs better than the baseline model.

Table 5. Accuracy of ensemble models.

Model	**Training Accuracy**	**Testing Accuracy**
Bagging	99.74%	75.89%
Adaboosting	92.36%	79.76%
XGboosting	90.32%	83.87%
Stacking	94.76%	82.65%

All of the ensemble algorithms produced better results and among them Xboosting gave us the best result (some selected results shown in Table 6). Figure 3 represents the performance metrics (Precision, Recall, and F-score) of 4 different ensemble models. Comparing the metrics, XGboosting has the highest precision (83.90), followed by Adaboosting (74.65), Stacking (80.32), and Bagging (72.78). XGboosting also has the highest recall (82.55), followed by Stacking (79.22), Adaboosting (73.24), and Bagging (71.5). However, when it comes to F-score, XGboosting has the lowest score (80.78), followed by Bagging (72.11), Adaboosting (72.76), and Stacking (80.32).

In our research, we observed that if the dataset is not small, it may not be necessary to apply boosting models. Boosting models are typically used when the dataset is small (as in our case) and the goal is to improve the performance of a weak learning algorithm. If a large enough dataset is available, other machine learning algorithms, such as deep learning models, could be used to classify spam SMS from Bengali text. Researchers could explore other techniques to improve the performance of the model, such as data augmentation, feature engineering, or transfer learning. However, if the dataset is still not big enough and re-boosting is not possible, the performance of the classification model may be impacted.

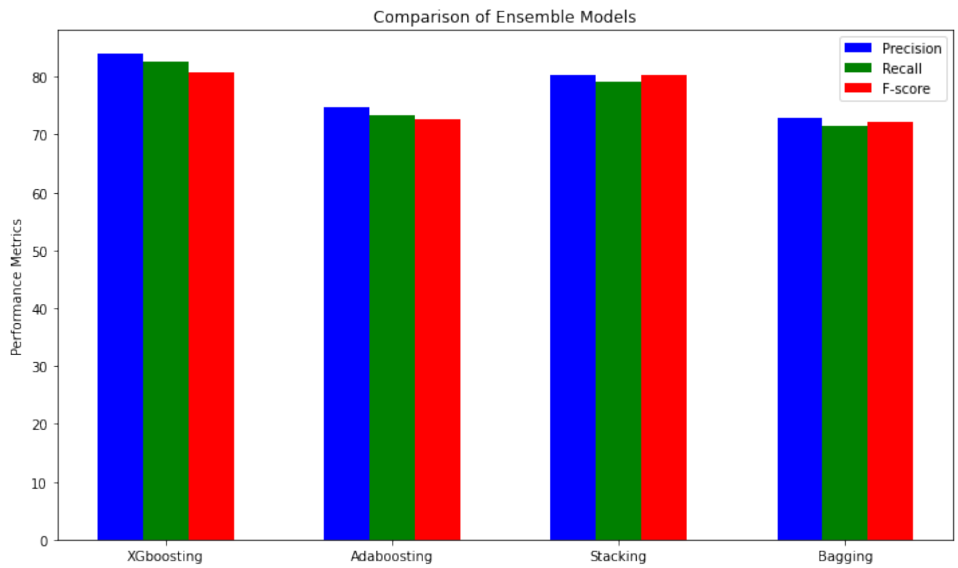

Figure 3. Precision, recall and F-score of ensemble models.

Table 6. True labels and predicted labels using XGboosting.

Data	English Translation	Actual Label	Predicted Label
মার্চ জুড়ে বাইক ও কার PROMO: DRL17 App:t.ly/sopi হটলাইন ০৯৬১১৯৯১১৭৭	Bike Car PROMO throughout March: DRL17 App:t.ly/sopi Hotline 09611991177.	SPAM	SPAM
আপনার ফোন দিয়ে প্রতিদিন 2500 আয় করুন:https://wa.me/639279003599	Earn 2500 daily with your phone: https://wa.me/639279003599.	SPAM	SPAM
জরুরি মুহূর্তে ১০০ টাকা পর্যন্ত ইমার্জেন্সি ব্যালেন্স পেতে ডায়াল *141*100#	Dial *141*100# to get emergency balance up to Tk 100	HAM	SPAM
আজই রিচার্জ করুন ২১২ টাকা । ৫ জিবি, ২০০ মিনিট এবং ৫০ SMS, মেয়াদ: ৩০ দিন:wa.me/8801917379501	Recharge today Rs.212 5 GB, 200 minutes and 50 SMS, validity: 30 days	HAM	HAM

5 Conclusion

This study aimed to classify and filter spam SMS in the Bengali language and has important implications for both business and cyber security. The use of bagging, boosting, and stacking ensemble models, along with preprocessing techniques, NLP modules, and the extraction of features using word N-grams, count vectorizer, and TF-IDF, resulted in high performance in detecting spam SMS. The combination of word N-grams and count vectorizer provided the best results. The boosting ensemble performed well in the task of detecting spam SMS in Bengali, and there is potential for further improvement through the use of deep learning models with a larger dataset. From a business perspective, the ability to accurately detect spam SMS can help organizations protect their customers from malicious activities and maintain the integrity of their communication channels. This, in turn, can lead to improved customer trust and brand reputation. From a cyber security perspective, the classification of spam SMS has the potential to prevent phishing scams, prevent the spread of malware and other harmful content, and protect sensitive personal and financial information. Our research will also have an impact on Bengali NLP, and future efforts will focus on the classification of multi-class concerns such as spam, ads, and ham SMS in Bengali text. The results of this study and the provided dataset have the potential to make a significant impact in both the business and cyber security domains.

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